

DATABRICKS THE UNIFIED DATA, ANALYTICS & AI PLATFORM



databricks

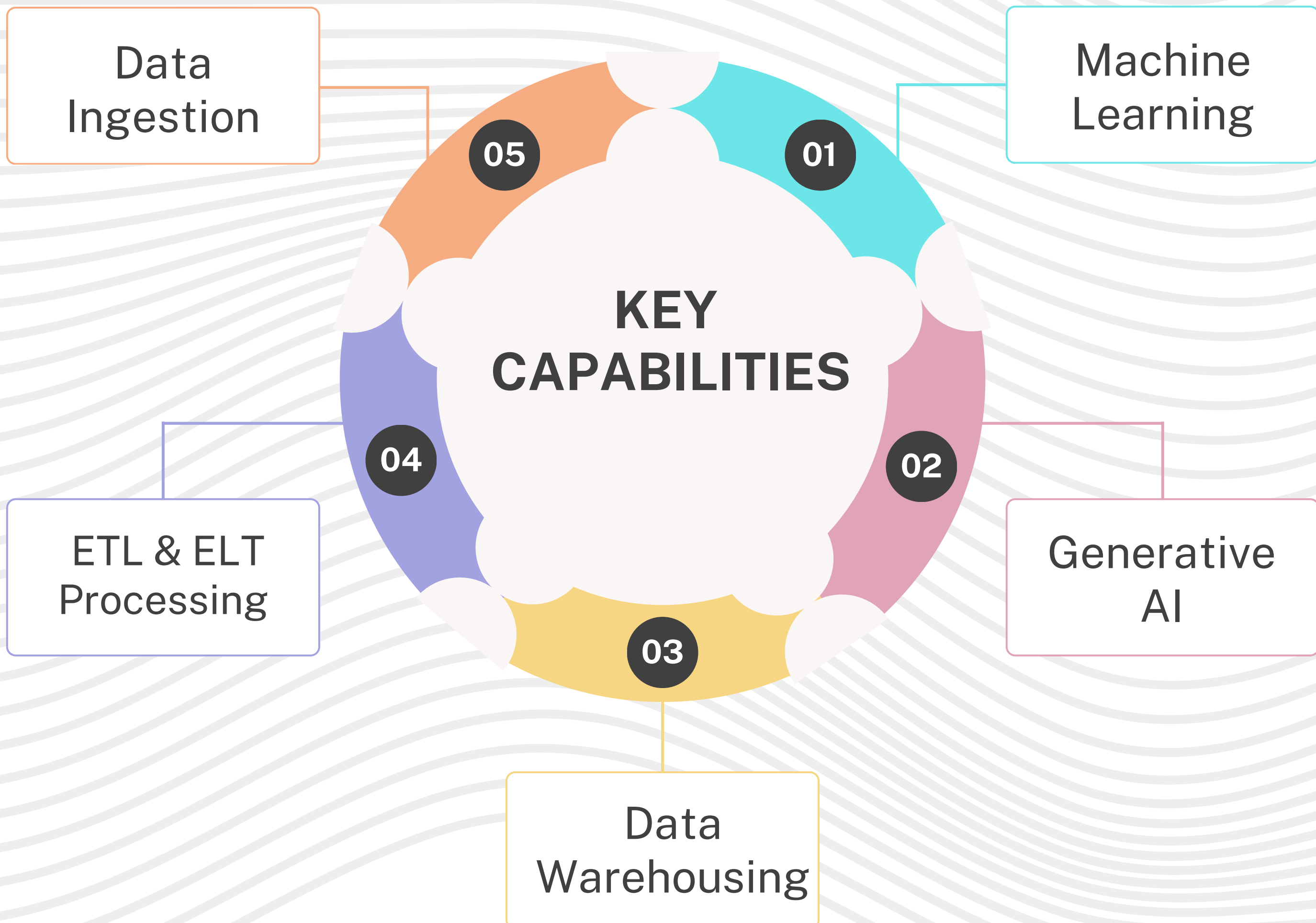
BY MICHAEL FERNANDES

WHAT IS DATABRICKS?

Databricks is a cloud-native platform created by the original creators of Apache Spark.

Instead of managing separate tools for storage, ETL, warehousing, and machine learning, teams can work together within one environment.

The goal is simple: turn raw data into business value faster.



THE PROBLEM DATABRICKS SOLVES

**Separate
Data Lakes**

**Separate Data
Warehouses**

**Separate ML
Platforms**

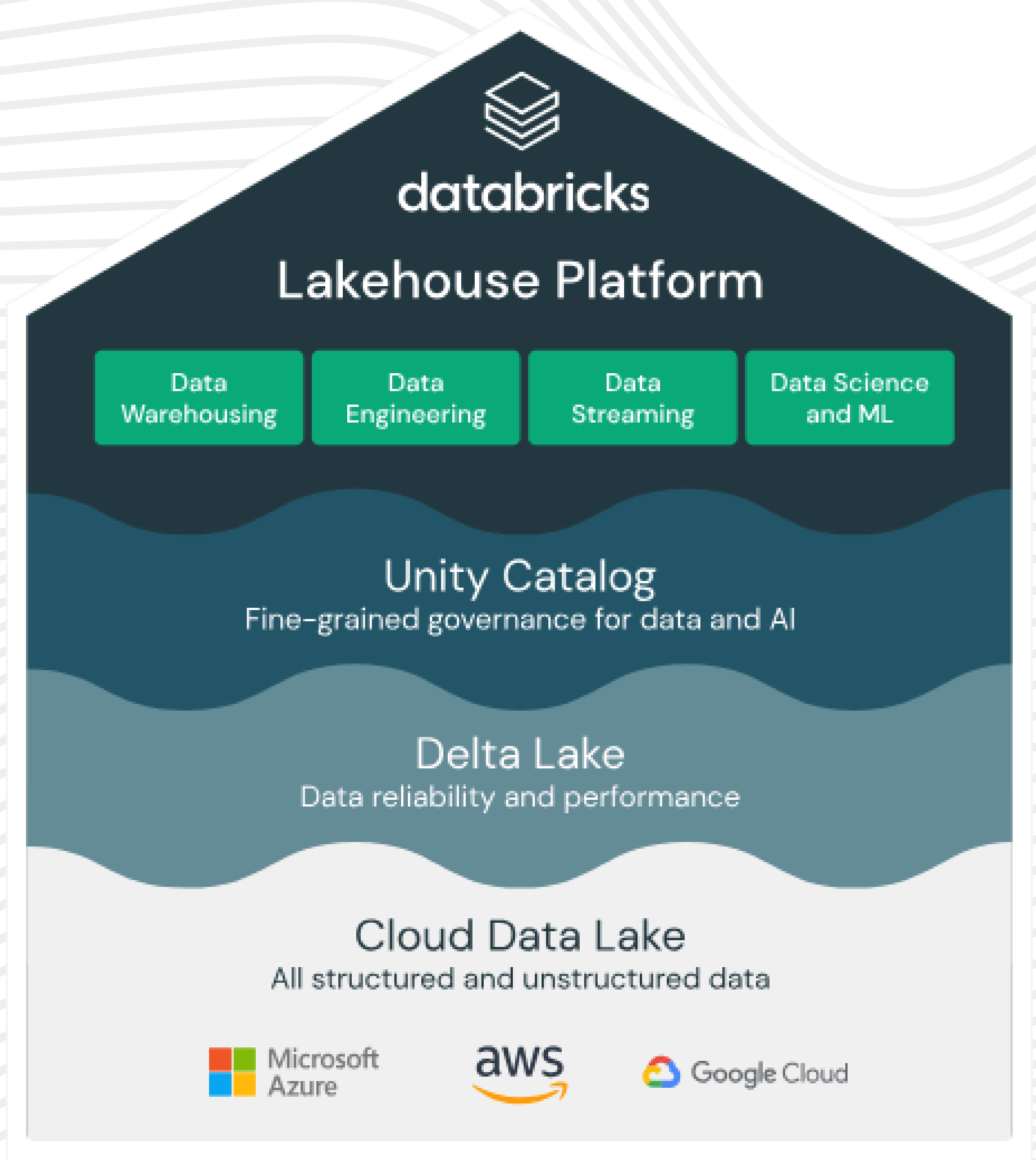
**Multiple
Data Copies**

**Governance
Challenges**

**High
Infrastructure
Costs**

Databricks eliminates these issues by bringing all workloads onto a unified platform where data teams collaborate using the same datasets.

LAKEHOUSE ARCHITECTURE



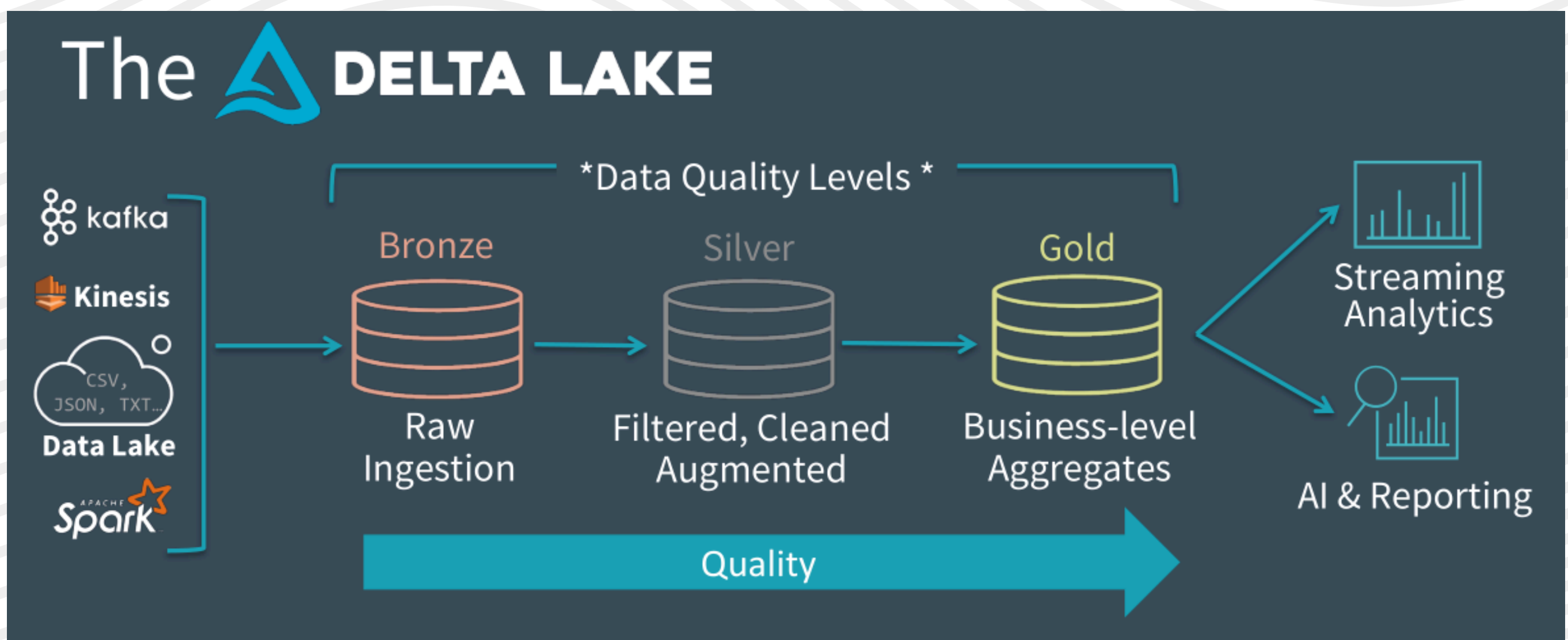
DELTA LAKE – THE FOUNDATION

Delta Lake is an open table format that brings database reliability to Data Lakes.

Traditional Data Lakes often struggle with data quality, consistency, and reliability.

Delta Lake solves these challenges by adding a transactional storage layer on top of cloud storage.

Delta Lake combines the scalability of a Data Lake with the reliability of a traditional database, making it the backbone of modern Lakehouse architectures.



MEDALLION ARCHITECTURE

The Medallion Architecture is Databricks' recommended framework for organizing data as it moves from raw ingestion to business-ready insights.

Instead of transforming data in a single step, data is refined through multiple layers, improving quality, governance, and reliability.

1. Bronze Layer — Raw Data

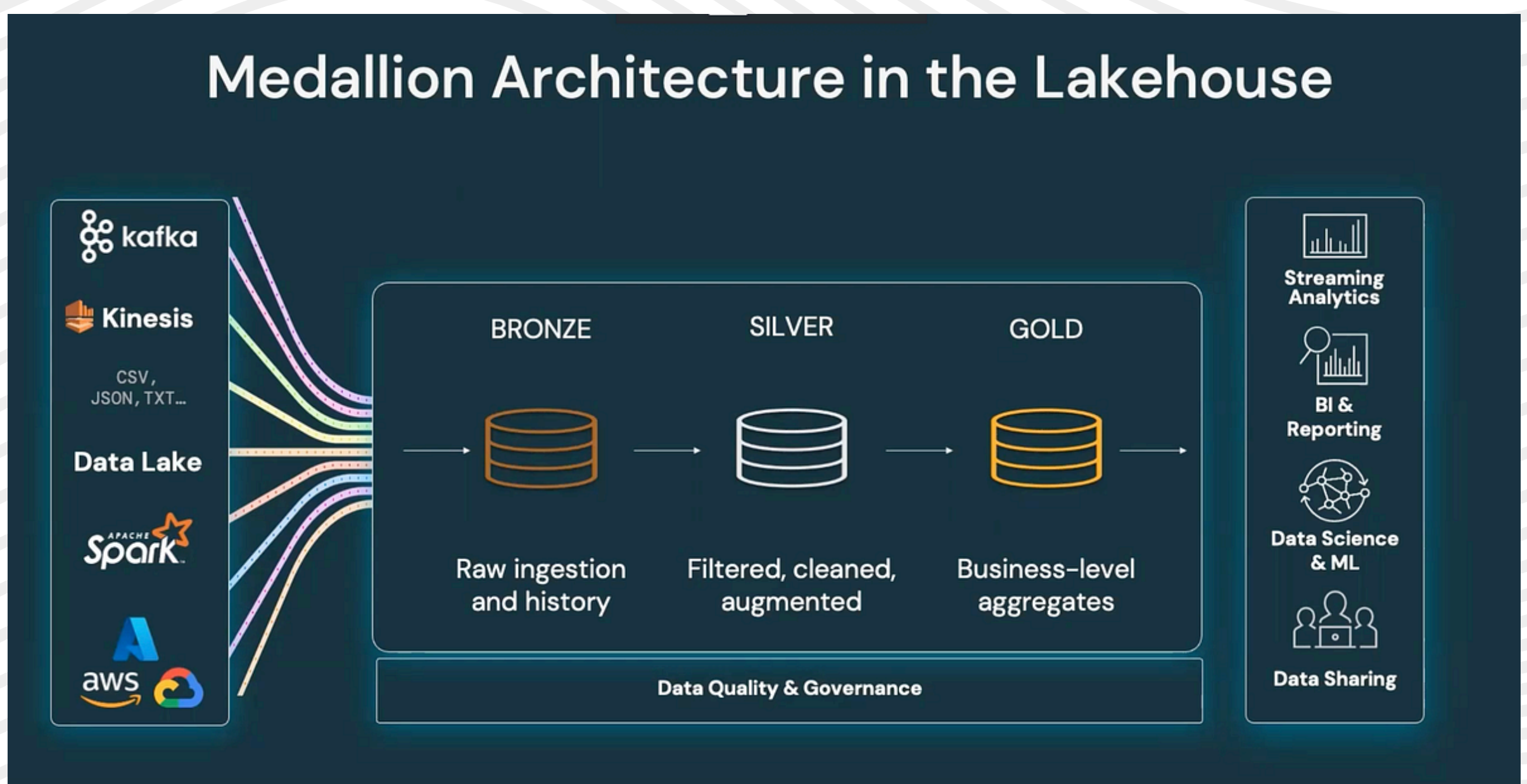
The Bronze layer stores data exactly as it arrives from source systems.

2. Silver Layer — Clean & Validated Data

Data is cleansed, standardized, and enriched in the Silver layer.

3. Gold Layer — Business-Ready Data

The Gold layer contains highly curated datasets optimized for business use.

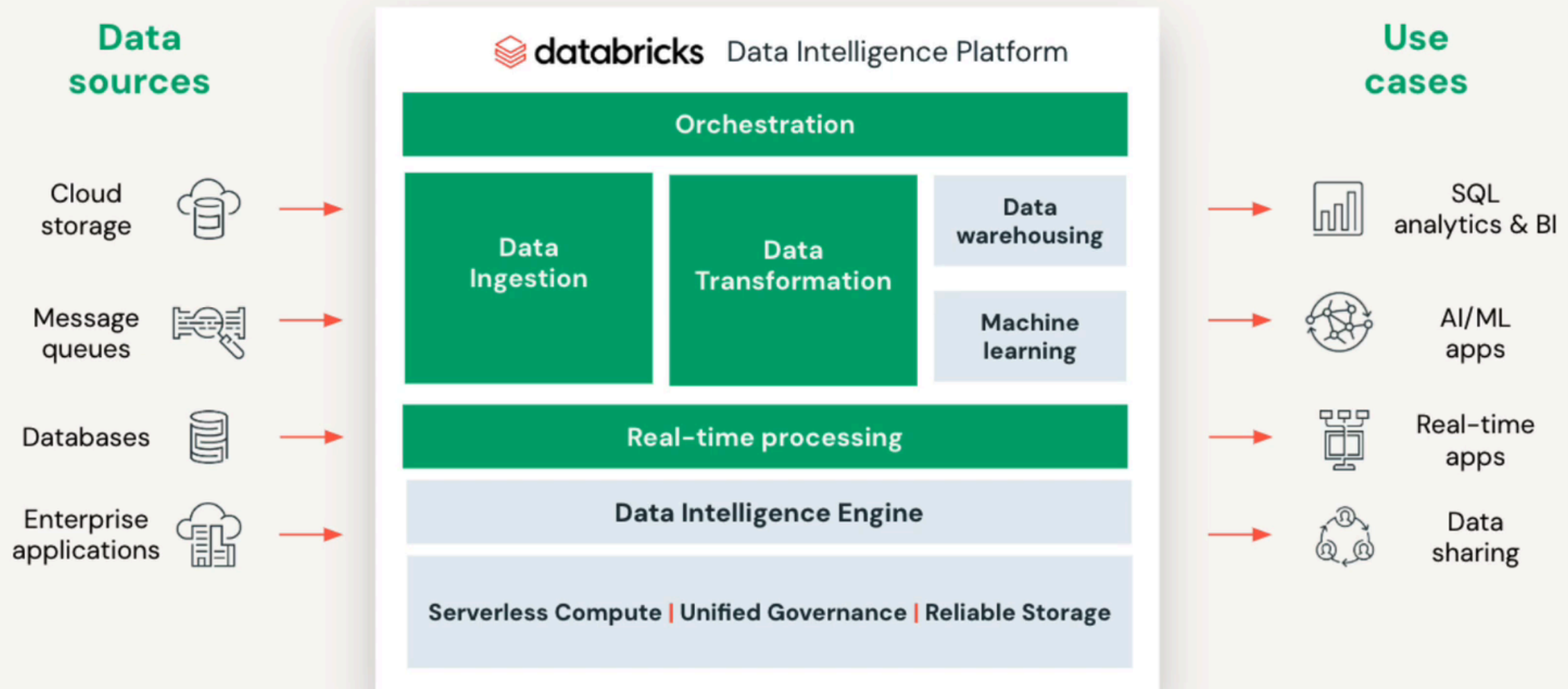


DATABRICKS FOR DATA ENGINEERING

Databricks has become one of the most widely used platforms for building modern data engineering pipelines at scale.

It combines Apache Spark, Delta Lake, Workflow Automation, and Cloud Storage into a single platform, enabling data engineers to process massive volumes of data efficiently.

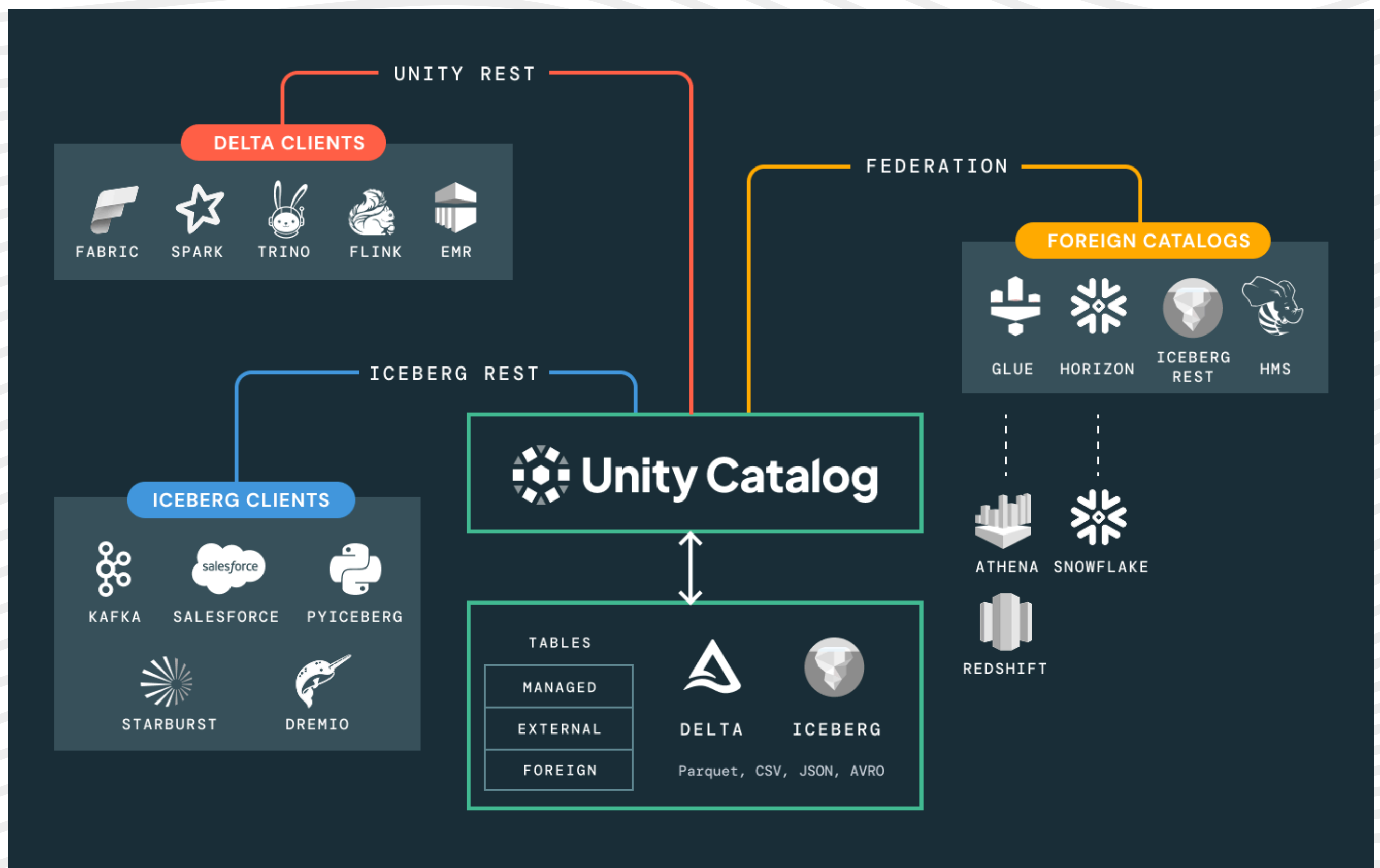
Data Engineering on Databricks



UNITY CATALOG & GOVERNANCE

As data grows, governance becomes critical. Unity Catalog provides centralized governance across all data assets.

Organizations can confidently manage thousands of datasets from a single control plane.



AI & MACHINE LEARNING ON DATABRICKS

Databricks is more than a data engineering platform—it's a complete environment for building, training, deploying, and managing AI and Machine Learning solutions.

**Feature
Engineering**

Model Training

**Experiment
Tracking
with MLflow**

**Model
Deployment &
Serving**

**Generative
AI**

**MLOps
Workflows**

WHY EVERY DATA ENGINEER SHOULD LEARN DATABRICKS

1. Master the Modern Data Stack

Databricks combines Spark, Delta Lake, Data Warehousing, Streaming, and AI in one platform. Learning it gives you exposure to the technologies used in modern data ecosystems.

2. Industry-Wide Adoption

Thousands of organizations across finance, healthcare, retail, and technology rely on Databricks. Its growing adoption has created strong demand for skilled professionals.

3. Handle Data at Massive Scale

Databricks is built to process terabytes and petabytes of data efficiently. It enables engineers to build high-performance pipelines without managing complex infrastructure.

4. Build End-to-End Pipelines

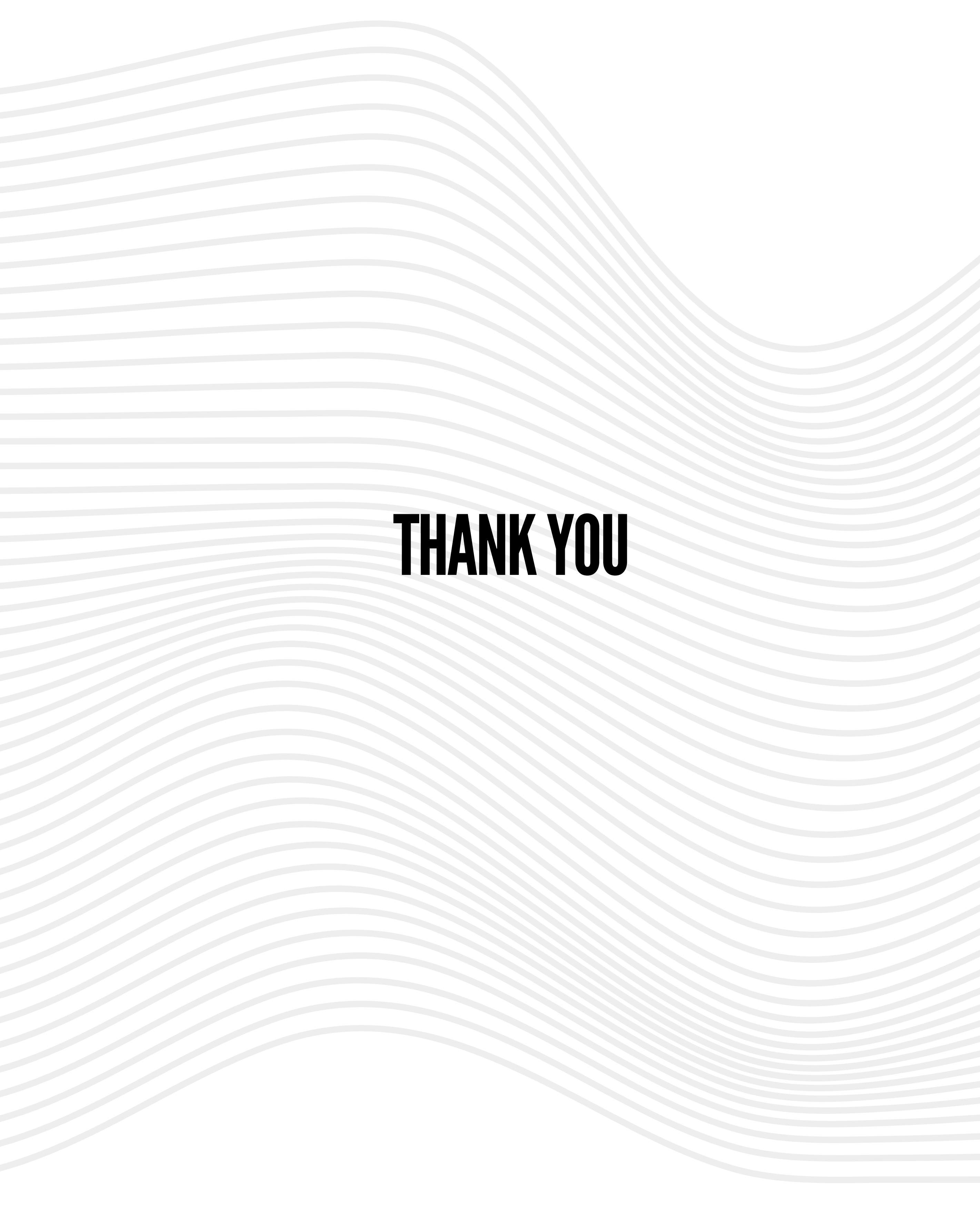
From data ingestion and transformation to analytics and AI, Databricks supports the entire data lifecycle. This reduces tool sprawl and simplifies development.

5. Understand Lakehouse Architecture

The Lakehouse approach is reshaping how organizations store and analyze data. Databricks is one of the leading platforms driving this architectural shift.

6. Boost Career Opportunities

Databricks skills are increasingly listed in job descriptions for Data Engineers, Analytics Engineers, and Data Platform Engineers. It has become a valuable addition to any data professional's toolkit.

The background of the image consists of numerous thin, light gray lines that flow horizontally across the frame. These lines are not perfectly straight but have a gentle, undulating wave pattern, creating a sense of movement and depth. The lines are more densely packed in some areas and more spread out in others, contributing to a textured, organic feel.

THANK YOU